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FIN 496
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Whitmore Capital Partners TAA Recommendations

Summary:

We recommend that Whitmore Capital Partners adopt the proposed SAA as the baseline with a market stress TAA overlay. In our backtest, the overlay improved the portfolio's annual return from 8.50% to 9.49% while reducing volatility from 10.41% to 9.13%. The TAA overlay also improved max drawdown from -30.54% to -19.05%. The tested TAA satisfied all three IPS performance criteria and risk constraints, while both benchmarks failed at least 2/3 of the stated constraints.

IPS Constraints:

The Whitmore Capital IPS requires an 8% annualized return, 15% volatility ceiling, 25% max drawdown tolerance, annual SAA rebalancing with 5 bps transaction costs. It also requires no cash drag and no shorting, as well as aggregate sleeve constraints with an amended 20% non-traditional sleeve cap. The proposed SAA and TAA overlay meets these requirements.

SAA Construction:

The SAA was constructed using the Mean-Variance (Markowitz) method with a Black-Litterman overlay for estimating the returns. Markowitz was chosen because Whitmore's objective is not just minimizing risk, but balancing return generation with the IPS volatility and drawdown limits. Black-Litterman was used to stabilize expected return estimates and reduce the sensitivity to noisy return inputs. The goal of this SAA is to provide returns within the IPS constraints during risk-on regimes. The SAA outperformed the 60/40 portfolio before adding the

TAA overlay, but still breached some of the IPS constraints, suggesting that a TAA overlay could add value.

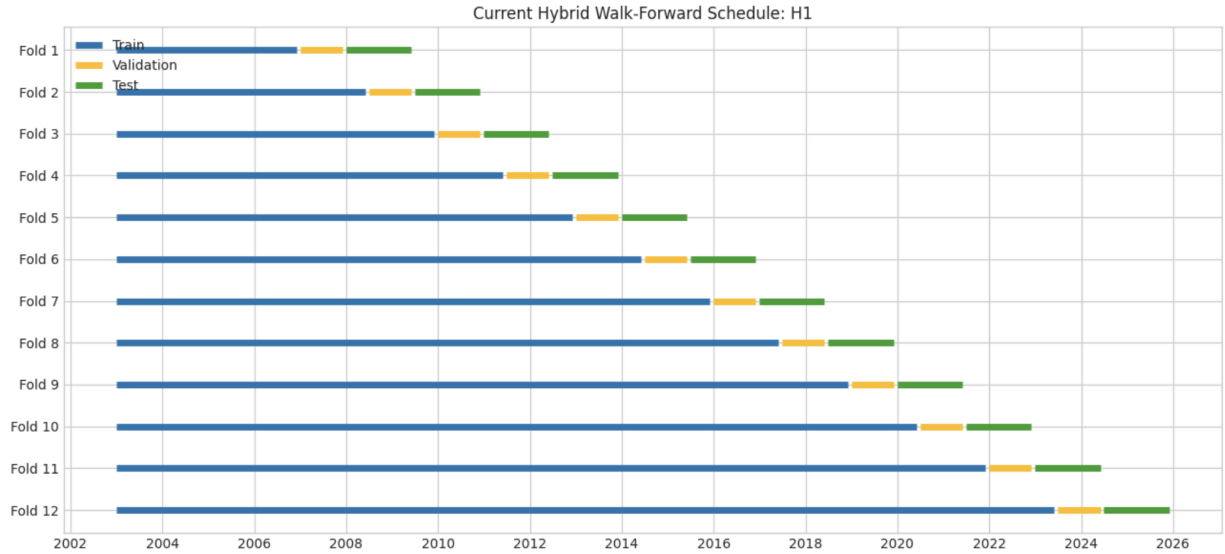
TAA Signal Rationale and Hypothesis:

The TAA signal was designed to predict times of market stress, allowing the portfolio to shift to a defensive position during risk-off regimes. The signals used are widening credit spreads (High Yield OAS), rising implied volatility (VIX), weakening equity trend (S&P 500), and the Russell 2000 and S&P 500 relative performance (RTY/ S&P 500). The rationale is that credit spreads and implied volatility act as forward-looking signals for market stress, while the equity performance is a confirmation of these signals. We hypothesize that when the forward-looking market stress signals deteriorate, expected returns on risky sleeves are reduced and downside risk rises faster than backward-looking macroeconomic data suggests. In these regimes, a TAA overlay should optimize away from riskier sleeves and more towards defensive and diversifying assets within the IPS constraints. The goal of the TAA is to improve risk-adjusted returns and reduce drawdowns. When triggered, the optimizer shifts away from equities (U.S. and international), REITs (due to the high correlation with equities), Bitcoin, and silver. These reductions should be dynamically shifted towards U.S. treasuries, TIPS, Gold, and the Japanese yen based on expected returns. By shifting the portfolio towards less risky assets, the volatility and drawdowns should be reduced, while relative performance improves.

Data, Walk-Forward Design, and Trial Disclosure:

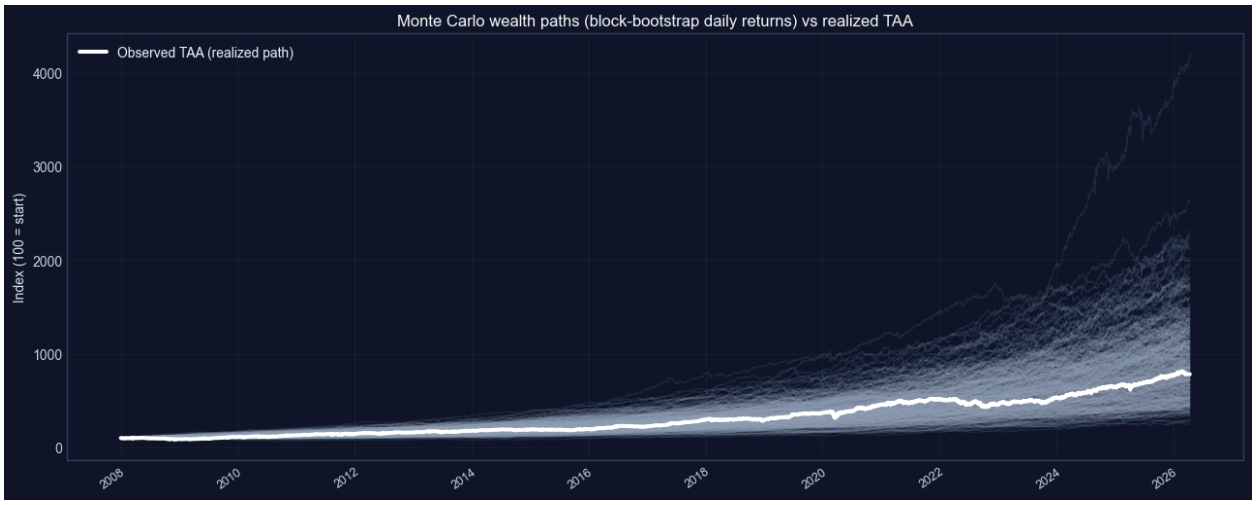
All data sourced came from one of two sources: the asset data was all sourced from the Zion database, fetched as far back as data was provided, and the signal data used was independently sourced using Bloomberg, also as far back as possible. Inevitably, this led to mismatches in data start times. The way this was handled was just to incorporate assets into the

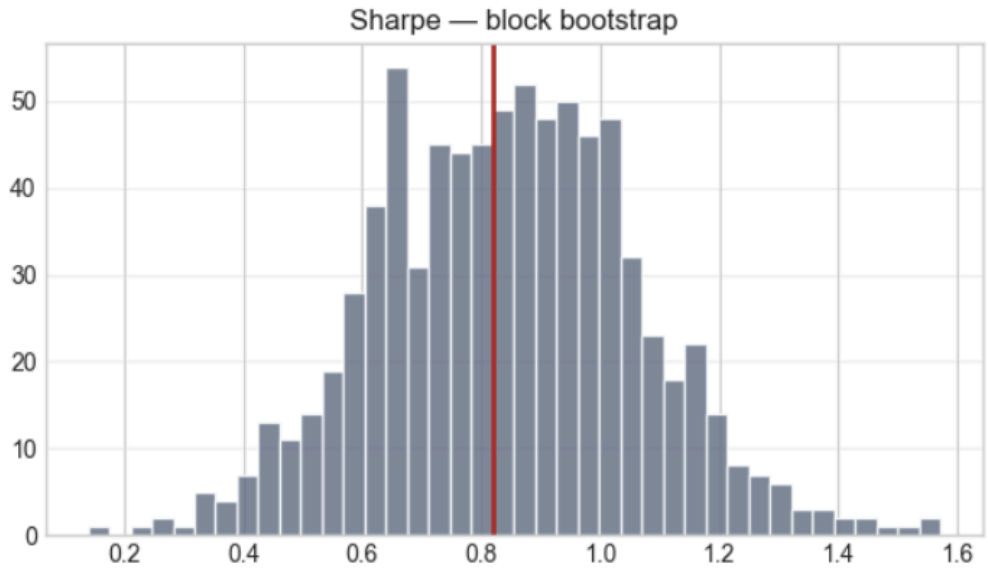
available asset universe as soon as it was made available. The reason behind the backtesting results starting in 2008 despite data going back to the 1970-1980 range is due to the walk-forward out-of-sample testing methodology implemented to prevent lookahead bias and overfitting. Testing was done using 12 train/val/test splits, with a 60-month window for training, 12 for validation, and 18 for testing. As seen in the chart below, these windows expand with each fold. 12 folds and rigorous data leakage prevention was used to ensure there was no way the implemented models could essentially "reward hack" to maximize the IPS objective, which would lead to impressive performance in theory but disastrous underperformance in practice. For each fold, an XGBoost model is fitted as a signal classifier, and an LSTM/Transformer model is fitted as a return prediction. Notably, they are only fit on the training set and applied on the val and test sets.



As for the trial disclosure, which delves deeper into the design of the training methodology, complexity begins to increase. An explicit grid search is run across the 4 available hypotheses/methodologies, and for each hypothesis, both an LSTM and Transformer model are fitted along with XGBoost, and for each model type, different asset setups are tested (including

ips_available, policy_core, policy_plus_bitcoin). Additionally, a hyperparameter sweep is conducted which would optimize seq_len, threshold, TAA band, risk aversion, BL tau, and regime_view_scale, among others. All of these results are saved to avoid selectively picking one performant model whilst discarding all others, more formally known as data snooping bias. They are saved with a status (keep/discard/crashed) which allows for easier display of top results. For additional security against this, a Monte-Carlo simulation of bootstrap-resampling on 500 random walks was run, as well as one run based on resulting Sharpe Ratio, both of which proving that the results attained by this model were by no means extraordinary at all.





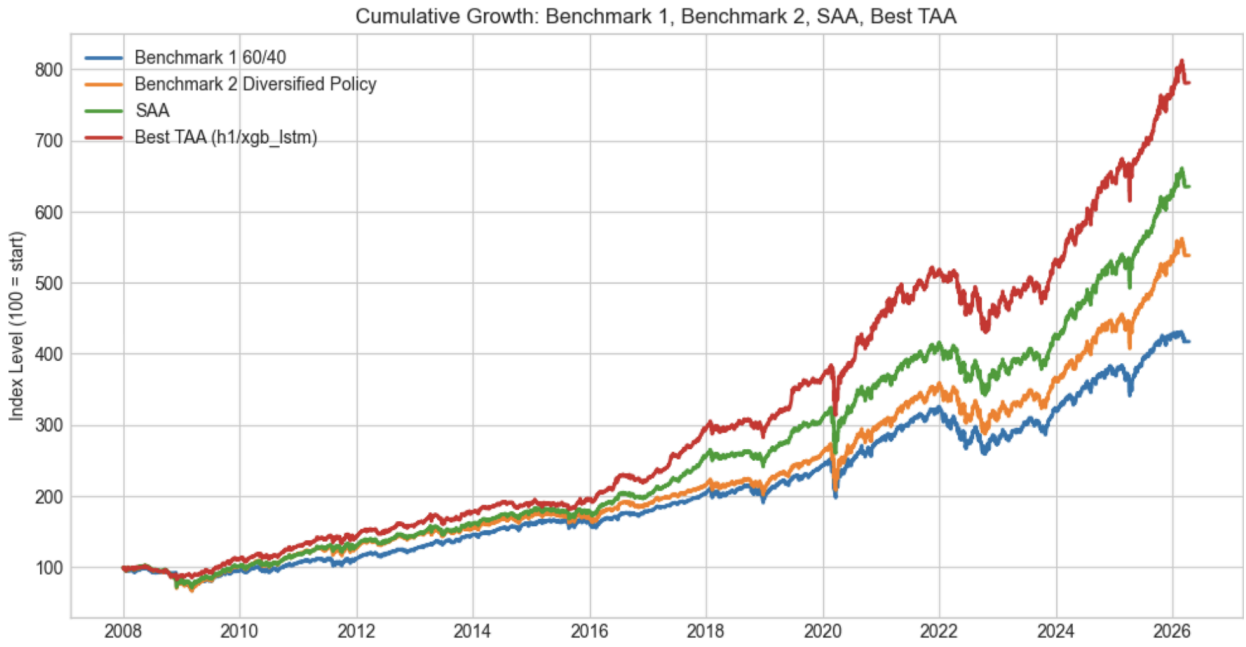
Something else to note is the long notebook and abundance of .md files. The long notebook is broken into 3 sections: the first of which does the hyperparameter sweep for the model decision itself (which does not need to be run again, due to the fact that the best model is already loaded into the artifacts subdirectory), the second of which loads in the backtesting modules, which used to exist as independent python scripts, and finally the chart generation at the very bottom, which is responsible for all of the results present in the report. The .md files are a summary of results from various model/hypothesis combos, and is skippable unless further investigation into alternative hypotheses and architectures is warranted, and the program.md used to run the autoresearch process that led to obtaining the best model in the first place, since it required more than a simple traditional grid search.

Performance:

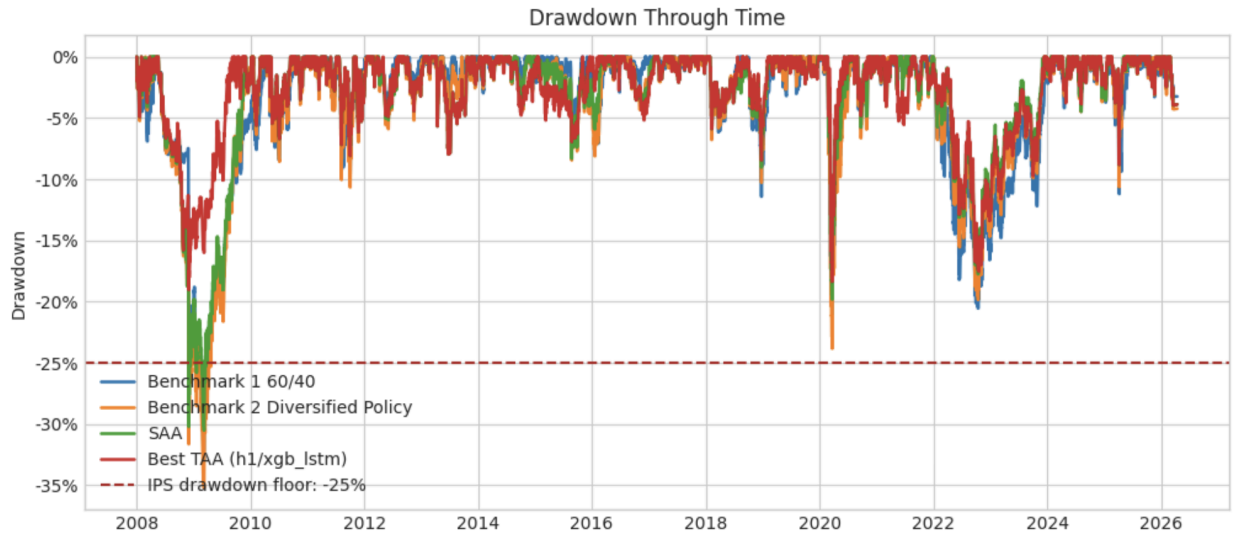
	ips_pass	total_return_pa	cumulative_return	volatility_pa	historical_var_95_daily	sharpe_rf2	calmar	max_drawdown
Benchmark 1 60/40	1/3	6.50%	3.171311	10.54%	-0.93%	0.462	0.197	-32.93%
Benchmark 2 Diversified Policy	1/3	7.71%	4.382456	11.91%	-1.03%	0.516	0.219	-35.24%
SAA	2/3	8.50%	5.351058	10.41%	-0.93%	0.645	0.278	-30.54%
Best TAA (h1/xgb_lstm)	3/3	9.49%	6.810608	9.13%	-0.85%	0.820	0.498	-19.05%

The results show that the market stress TAA overlay added value to the SAA portfolio and significantly outperformed the benchmarks. The 60/40 portfolio generated a 6.50% annualized return with a 10.54% volatility and a -32.93% max drawdown. The diversified policy benchmark improved on the basic 60/40 by improving the return to 7.71%, still failing to meet the 8% IPS return target, and had the worst drawdown at -35.24%. The SAA outperformed both of the benchmarks with a return of 8.50% and reduced volatility of 10.41%, but still breached the maximum tolerated drawdown as specified by the IPS. The TAA overlay further improved the SAA, producing the strongest result. It increased annualized return to 9.49% (net of the 5 bps transaction cost), well above the 8% goal, while also reducing volatility to 9.13%. The TAA's

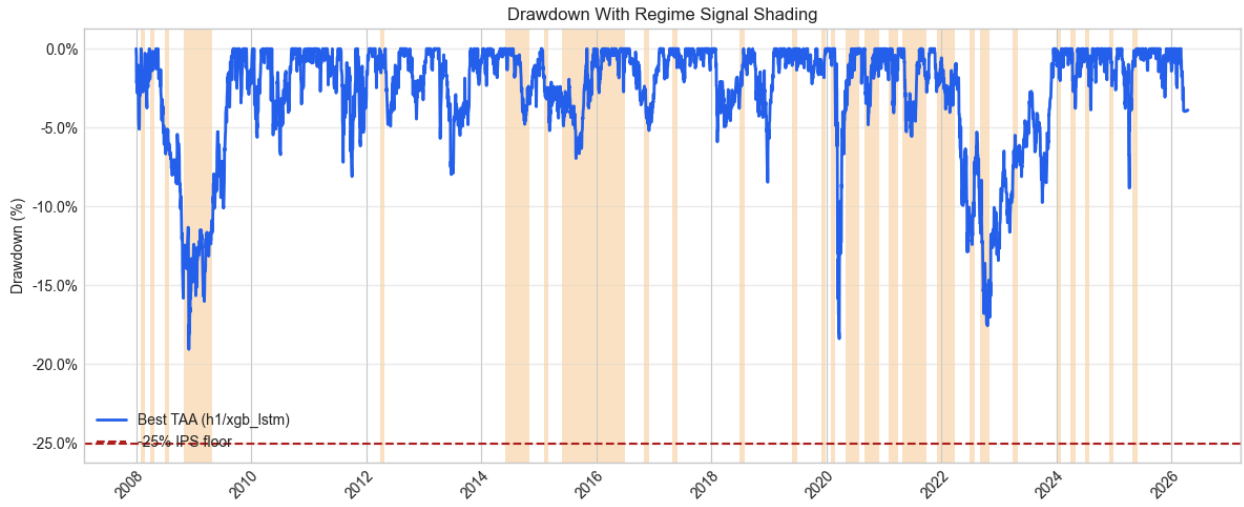
max drawdown was the only one to pass the IPS constraints, with a max drawdown of -19.05%. The overlay also significantly improved risk adjusted metrics, with a Sharpe ratio of .82, and a Calmar ratio of .498. This indicates that the TAA did not increase returns by taking more risk, rather it improved the portfolio's risk-adjusted returns by reducing downside exposure during stress periods.



The cumulative growth chart shows the material outperformance of the TAA portfolio, especially after 2016 when the four portfolios diverge more significantly. The TAA maintains a much stronger growth rate through stress periods, especially following the COVID selloff. The TAA offered less protection during 2022, when equity and bond markets sold off in tandem and traditional diversification benefits weakened, although this affected all of the portfolios.



An exploration of drawdowns reveals the main source of value for the TAA, which is significantly reduced compared to the other portfolios. This shows how the TAA's primary contribution was not just higher return, but better downside protection. With this protection, the portfolio is able to more effectively compound over time by avoiding short term losses.



The TAA overlay triggered during times of market stress, allowing the portfolio to reduce the downside risk in a timely manner, and allowed the portfolio to recover faster than the SAA and 60/40 portfolios, especially in 2009.

SAA and TAA Contribution Analysis:

The SAA contributed most of the returns for the portfolio. Relative to the benchmark portfolios, the SAA improved annualized return and risk-adjusted returns by strategically diversifying across the sleeves rather than relying only on the traditional 60/40 mix, or the diversified benchmark. However, the SAA alone did not solve the portfolio's downside problem, as it still breached the maximum acceptable drawdown. This suggests that strategic diversification added value, but was not sufficient to meet all of Whitmore's return and risk constraints. The TAA overlay improved the SAA's value by reducing downside risk during market stress. Relative to the SAA, the TAA overlay increased annual return from 8.50% to 9.49% while reducing volatility from 10.41% to 9.13%. The TAA was able to materially improve the returns by reducing the max drawdown from -30.54% to -19.05%, allowing for better risk-adjusted returns (Sharpe and Calmar). The main contribution of the TAA was regime sensitivity. When market stress indicators deteriorated, the portfolio reduced exposure to risky sleeves, allowing the portfolio to better retain value during market stress, therefore improving compounding over time.

Recommendation & Limitations:

We recommend that Whitmore Capital Partners adopt the proposed SAA as the baseline, along with the market-stress TAA overlay for conditional use because it materially improved nominal and risk-adjusted returns, reduced volatility, and significantly improved max drawdown in the backtest, net of transaction costs. Relative to the SAA, the TAA increased annualized return from 8.50% to 9.49%, reduced annualized volatility from 10.41% to 9.13%, and reduced max drawdown from -30.54% to -19.05%. This gave the TAA significantly better risk-adjusted metrics, and was the only portfolio that was firmly within the IPS constraints over the tested period. The overlay is an effective risk management and risk-adjusted return enhancement, not a

return-maximizing strategy. The TAA adds the most value during market stress and regime deterioration by preserving capital and improving recovery, allowing for stronger compounding over time. The TAA remains fully governed by the IPS constraints, including sleeve caps, no shorting or cash drag, and the amended 20% non-traditional cap. We recommend adopting the SAA with the TAA overlay with monthly monitoring of TAA attribution relative to the SAA baseline, consistent with the IPS governance framework.

The overlay is most likely to add value when all of the signals deteriorate at the same time, which is likely during prolonged periods of market stress, when the markets are repricing risk rather than experiencing a brief sentiment shock. The TAA is especially useful when trying to limit drawdowns without abandoning return targets, making it well suited for Whitmore's goals. However, the overlay may struggle when the market sees sharp and quick volatility in the signals, like when stress indicators spike briefly before quickly reversing. This is because the signal may trigger after a violent initial selloff before quickly rebounding, meaning there is a risk of the overlay shifting too late or too long. Results also depend on the quality, timeliness, and availability of signal data, increasing the risk of false positives due to poor data quality, or when volatility spikes without broader signal confirmation. Therefore, the TAA should be viewed as a decision support tool rather than a foolproof system that removes the need for careful oversight. Still, the backtest suggests that it can be a robust and effective tool for downside protection and smoother compounding. With these considerations in mind, we still recommend that the SAA with the TAA overlay be adopted by Whitmore Capital Partners as it meets the client's needs and constraints effectively with comfortable margins.